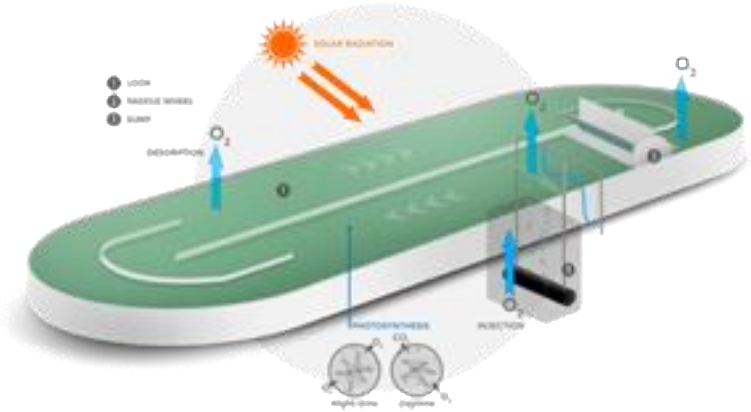
IFAC World Congress 2026

**BUSAN**

# Benchmark Challenge: Process Control Research of Outdoor Microalgae Raceway Reactors

## User handbook



**Organization:**

# Process Control Research of Outdoor Microalgae Raceway Reactors
# IFAC World Congress Control Benchmark Challenge – 2026

## Organized by

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Automatic Control, Robotics and Mechatronics Research Group
University of Almería (UAL), Spain
https://arm.ual.es/arm-group/

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## In collaboration with:

IFAC Technical Committee 9.4 on Control Education
IFAC Subcommittee on Benchmark Control Problems
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# Introduction

Outdoor microalgae raceway reactors are among the most cost-effective technologies for large-scale algal biomass production and carbon capture. Their open configuration allows direct use of solar energy and ambient resources, but at the cost of strong exposure to environmental disturbances, including diurnal radiation cycles, temperature fluctuations, wind, and humidity. Under these conditions, maintaining key process variables, such as pH, dissolved oxygen (DO), and culture temperature, within suitable operating ranges is essential to ensure stable operation and to keep biomass production close to optimal conditions.

From a control engineering perspective, outdoor raceway reactors constitute a challenging multivariable process that combines physicochemical equilibria, biological growth kinetics, gas–liquid mass transfer, and thermal energy balances. Because of the underlying biological, chemical, and thermal couplings, process variables exhibit highly nonlinear, tightly coupled, and time-varying dynamics. Actuator constraints, transport delays, and strong day-night regime switching further complicate the design and evaluation of advanced control strategies. Despite significant progress in modeling and control of microalgae systems, the lack of open, reproducible, and high-fidelity benchmark problems has limited fair comparison between alternative control methodologies.

This benchmark challenge addresses this gap by providing a unified and openly available simulation platform for the evaluation of process control strategies applied to outdoor microalgae raceway reactors. The benchmark is based on a high-fidelity dynamic model, experimentally calibrated using data from a real pilot-scale raceway reactor operated at the CIESOL research facilities located at the IFAPA research center (Almería, Spain). The model captures the essential biological, chemical, thermal, and hydraulic mechanisms governing reactor behavior under realistic outdoor conditions.

In the context of the IFAC World Congress, this benchmark is proposed as a standardized control challenge aimed at fostering methodological comparison, reproducibility, and educational value. Participants are invited to design and test control algorithms under identical operating scenarios, disturbance profiles, and performance metrics. The benchmark supports both classical control approaches (e.g., PI/PID) and advanced strategies (e.g., predictive, economic, or learning-based control), enabling objective performance assessment across a wide range of control paradigms.

This handbook provides a concise and practical guide to the benchmark, summarizing the system description, control objectives, performance evaluation criteria, and submission procedure. Detailed modeling equations and implementation aspects are available in the reference article, while the complete simulation environment and baseline controllers are provided through the official GitHub repository.

## Benchmark Philosophy

The philosophy of this benchmark is to provide a realistic, reproducible, and methodologically neutral platform for the evaluation of control strategies applied to outdoor microalgae raceway reactors. Rather than favoring a specific control paradigm, the benchmark is designed to expose participants to the intrinsic challenges of the process, strong environmental disturbances, nonlinear multivariable interactions, actuator limitations, and day–night regime switching, under identical and fully specified operating conditions. Performance assessment is therefore based on a common simulation environment, fixed disturbance scenarios, and unified performance indices, ensuring fair and transparent comparison across classical, predictive, economic, and data-driven control approaches. By focusing on regulation of key variables directly linked to biological productivity and operational robustness, the benchmark aims to bridge control theory, process engineering, and real-world outdoor bioprocess operation.

# Objectives

The benchmark is designed so that students and researchers can focus on control design and performance evaluation, without needing to modify the underlying process model. The main control objective is to keep the reactor operating under favorable conditions for biomass production, despite strong environmental disturbances. This is achieved by regulating a set of key process variables using the available actuators provided in the benchmark. More specifically, the benchmark has the following control objectives:

**1) Regulation of key process variables**

Participants are required to design controllers to regulate:

- pH,
- DO,
- Culture temperature,

so that these variables remain close to their prescribed reference values. Maintaining these variables within suitable operating ranges is essential to ensure stable reactor operation and to promote high biomass productivity, which is the overall goal of the benchmark.

**2) Disturbance compensation**

The control strategies should be able to cope with time-varying environmental disturbances, including:

- Solar radiation,
- Ambient temperature,
- Wind speed,
- Relative humidity.

These disturbances directly affect biological activity, gas transfer, and thermal behavior, and therefore represent a key challenge of the benchmark.

Moreover, dilution and harvesting operation act as measurable disturbances with relevant impact on some of the process control loops, mainly, pH and DO control.

**3) Efficient use of actuators/resources**

In addition to tracking performance, controllers are expected to use the available actuators efficiently. This includes avoiding unnecessary or excessive actuation of:

- $CO_2$ injection (pH control),
- Air injection (DO control),
- Thermal actuation (temperature control).

This objective reflects practical operational constraints and encourages realistic control solutions.

**4) Fair and reproducible comparison**

All participants work with:

- The same high-fidelity process model,
- Identical disturbance scenarios,
- The same performance indices.

This ensures that different control strategies can be fairly and objectively compared, based solely on their control performance.

**5) Methodological openness**

The benchmark does not impose any specific control methodology. Participants are free to implement classical controllers (e.g., PI/PID), Predictive and economic control strategies, advanced or learning-based approaches, as long as they comply with the provided interfaces and constraints.

# System description

The benchmark is based on a high-fidelity dynamic model of an outdoor microalgae raceway reactor, specifically developed for control research and education. A complete and detailed description of the process, including the underlying model equations, parameter calibration, and baseline control schemes, is provided in the reference paper available on

https://arxiv.org/abs/2512.15916

Users interested in a deeper understanding of the biological, chemical, thermal, and hydraulic dynamics of the reactor are strongly encouraged to consult the article and references therein.

The modeled system represents a real pilot-scale raceway reactor operated at the CIESOL research facilities (Almería, Spain). The reactor is exposed to realistic outdoor conditions and is affected by time-varying meteorological disturbances such as solar radiation, ambient temperature, wind speed, and humidity. These disturbances strongly influence photosynthesis, gas–liquid mass transfer, carbonate chemistry, and heat exchange, resulting in nonlinear and highly coupled system behavior.

Figure 1 summarizes the three coupled sub models:

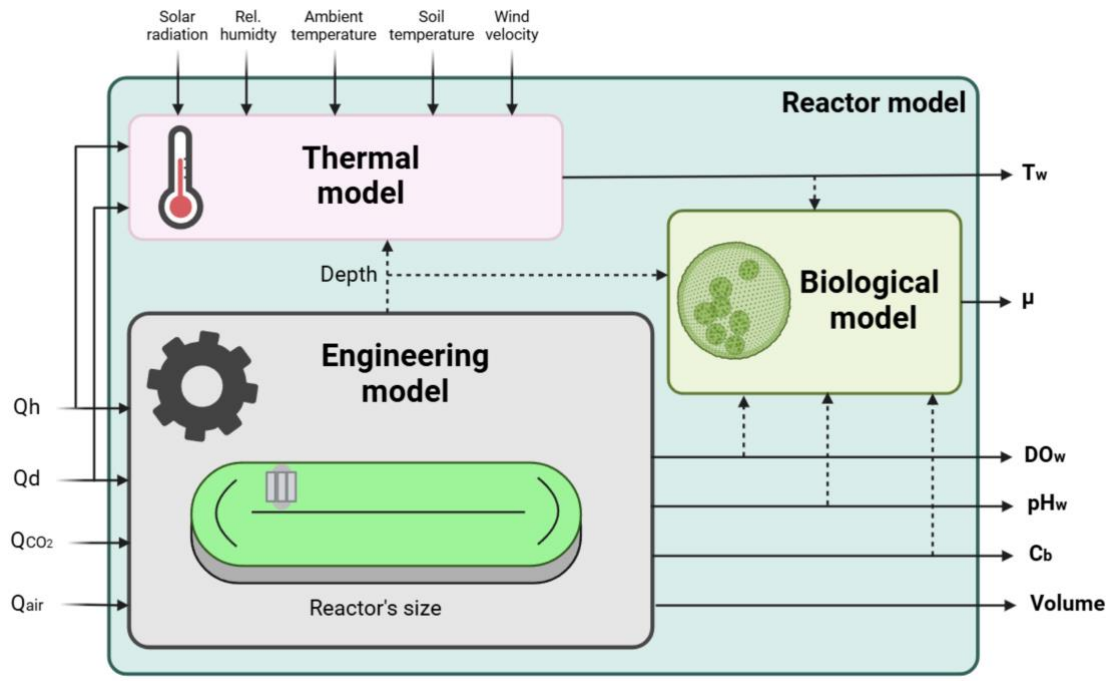


*Figure 1 - Architecture of the raceway benchmark model.*

From a control perspective, the raceway reactor combines several interacting subsystems:

1. Thermal sub model: describes the bulk temperature dynamics including meteorology, evaporation, conduction, convection and the spiral heat exchanger.
2. Biological sub model: computes light, temperature, pH and DO limitation indexes, photosynthesis reaction, growth rate μ and maintenance m.
3. Engineering (macroscopic) sub model: implements, the volume, biomass, total inorganic carbon, cation and $O_2$ balances, together with gas transfer and carbonate speciation.

For the purpose of this benchmark challenge, users do not modify the process model itself. Instead, interaction with the system is performed exclusively through well-defined control interfaces provided in the MATLAB simulation environment. This allows participants to focus on controller design, tuning, and performance evaluation, while working with a realistic and experimentally validated process model.

A simplified view of the control problem is therefore adopted: the process model, disturbance scenarios, and simulation settings are fixed, while the control algorithms implemented by the user determine the overall performance of the closed-loop system.

# Benchmark Control Structure

The benchmark is organized around a set of coupled single-loop control problems, each one associated with a key process variable and a specific actuator. Although each control loop can be designed independently, all loops interact through the underlying biological, chemical, and thermal dynamics of the raceway reactor.

Participants interact with the benchmark by implementing control laws that compute the actuator signals at each simulation step. The process model, disturbance scenarios, and simulation framework are fixed, while the controller logic is entirely user-defined within the provided MATLAB functions.

For this benchmark challenge, three control loops are considered active:

- pH control,
- DO control,
- Temperature control.

The harvesting and dilution strategy is fixed and varies only according to a predefined daily schedule. As a result, users do not design controllers for harvest or dilution but must account for their effect as an external disturbance on the system, such as described above.

## Control Objectives and Control Loops

Each control loop is defined by a controlled variable, a manipulated variable, and a fixed reference value. These loops operate continuously over multi-day outdoor scenarios and are subject to strong diurnal variations caused by solar radiation and ambient conditions.

### pH Control

Controlled variable: pH

Manipulated variable: $CO_2$ injection flow rate

Reference value: pH = 8.0

The pH dynamics are strongly affected by photosynthesis and respiration, which vary significantly between day and night. During daylight hours, photosynthetic activity increases pH due to $CO_2$ consumption, while at night respiration causes pH to decrease. The $CO_2$ injection controller must therefore continuously compensate for these effects to maintain pH close to its reference.

## Dissolved Oxygen (DO) Control

Controlled variable: Dissolved oxygen (DO), expressed as percentage of saturation

Manipulated variable: Air injection flow rate

Reference value: DO = 150%

DO exhibits pronounced diurnal behavior: it typically increases during the day due to photosynthetic oxygen production and decreases at night due to respiration. The air injection controller is responsible for preventing excessive oxygen accumulation during the day while ensuring sufficient oxygen levels during nighttime operation.

## Temperature Control

Controlled variable: Culture temperature

Manipulated variable: Water flow rate through the sump-mounted heat exchanger (and inlet temperature, when applicable)

Reference value: Temperature = 30 °C

Culture temperature is influenced by solar radiation, ambient temperature, wind, evaporation, and thermal inertia. During the day, solar heating can raise the culture temperature above the desired value, while nighttime conditions may lead to cooling. The temperature controller must regulate heat exchange to keep the culture temperature close to its reference despite these daily fluctuations.

## Harvest and Dilution (Fixed Strategy)

Harvesting and dilution are not controlled by the participant in this benchmark. Instead, they follow a predefined schedule:

On one simulated day, harvesting occurs at 09:00,

On the following day, harvesting occurs at 12:00,

with this pattern repeating over the simulation horizon. This strategy introduces step changes in culture volume and biomass concentration, which act as known but unavoidable disturbances that the controllers must handle.

This control structure allows participants to focus on regulation, disturbance rejection, and actuator efficiency, while working with a realistic and dynamically coupled outdoor process.

# Controller Implementation in MATLAB

The benchmark is designed so that participants can focus exclusively on controller design, without modifying the underlying process model or simulation framework. All core model files and simulation routines are protected, ensuring fair comparison and full reproducibility of results.

Participants must run the code:

- `Benchmark_main.m`

## What participants are allowed to modify

Participants must implement their control strategies only by editing the controller functions provided in the player/ folder. These functions are called automatically by the main simulation script at each simulation step. In the main code, participants can found the call for their functions in the lines: as depicted in Figure 2.

```
%% ---------------- Controllers (one per actuator) ----------------
ctrl = struct();
ctrl.fn_pH_CO2  = @controller_pH;
ctrl.fn_DO_air  = @controller_DO;
ctrl.fn_Temp_HX = @controller_Temp_HX;
```

*Figure 2 - Functions call in Benchmar_main.m*

The default controller files are:

- `controller_pH.m` – pH control via $CO_2$ injection
- `controller_DO.m` – DO control via air injection
- `controller_Temp_HX.m` – temperature control via heat exchanger

In the versions available on GitHub, simple On/Off controllers are implemented for pH and DO, while no active control is applied to the temperature loop. These files are intended as reference implementations that illustrate the required structure and data flow of the controllers.

Participants are encouraged to use these files as a starting point to develop their own control strategies. In particular, they may:

- Design custom control logic inside the pH, DO, and temperature controller functions (e.g., start/stop conditions, rule-based decisions, time-dependent behavior, or use of environmental information).
- Create new controller functions implementing alternative approaches (e.g., PI, PID, MPC, EMPC, rule-based or learning-based controllers), provided that the function interfaces are preserved, and the corresponding function names are correctly specified in `Benchmark_main.m`.

This modular structure allows participants to experiment freely with different control concepts while ensuring compatibility and fair comparison within the benchmark framework.

## Controller function interface

Each controller function (.m) must follow the exact input/output structure defined in the repository, as documented in the GitHub README, in this link:

https://github.com/guzmanjl/benchmarkmicroalgae

In resume, all controller functions in the benchmark follow a common input–output structure, which allows different control strategies to be tested under identical conditions. A typical controller function has the following form:

```
[st_CtrlSignals, state] = controller_pH_PI( ...
    Timeline, obs, refs, env, future, st_CtrlSignals, state)
```

At each simulation step, this function is called by the benchmark engine. The controller receives the current system information, computes the control action, and returns the updated control signals.

The controller function receives the following inputs:

- Timeline
  Provides time-related information such as the sampling time, current simulation time, and position within the day (hour and minute). This allows the controller to implement time-based logic, scheduling, or day-night dependent strategies.

- obs (observations)
  Contains the measured or estimated process variables at the current time step, including pH, DO, temperature, biomass concentration, and culture depth. These are the main signals used for feedback control.

- refs (references)
  Contains the reference values for the controlled variables (pH, DO, and temperature). Controllers are expected to regulate the process so that the observed variables track these references.

- env (environmental variables)
  Provides the current environmental conditions affecting the reactor, such as solar radiation, ambient temperature, wind speed, and humidity. These signals can be used for feedforward compensation or adaptive control strategies.

- future (optional preview information)
  Contains future trajectories of environmental disturbances when available. This information enables the implementation of predictive or anticipative control strategies, such as MPC or rule-based scheduling.

- st_CtrlSignals (control signal structure)
  A shared structure that collects all control commands generated by the different controllers ($CO_2$, air, harvest, dilution, heat exchanger). Each controller is expected to update only its corresponding control signal, leaving the others unchanged.

- state (controller state)
  A persistent structure used to store controller parameters and internal states (e.g., integrator values, filters, model parameters). This structure is preserved between simulation steps, allowing implementation of dynamic controllers.

The controller function must return:

- st_CtrlSignals
  The updated control signal structure. For example, a pH controller modifies only the $CO_2$ injection command (Qco2), while leaving all other fields unchanged.

- state
  The updated controller state, including internal variables such as integrator states or adaptive parameters.

**Important remarks for participants**

Controllers must respect the provided function interface to ensure compatibility with the benchmark. Actuator limits, signal saturation, and transport delays are handled internally by the simulator, not by the controller. However, Participants are encouraged to consider these restrictions in their own controller.

Participants are free to implement any control logic inside the controller functions, as long as inputs and outputs follow the specified structure.

This standardized interface ensures that all controllers interact with the process in the same way, enabling fair comparison and reproducible evaluation of different control strategies.

## Graphics and Visualization of Results

After each simulation run, the benchmark automatically generates a set of graphical outputs that allow participants to visually evaluate the behavior of their controllers. These plots are produced by the provided visualization scripts and require no additional user intervention.

The figures are intended to help participants answer three key questions:

- Are the controlled variables tracking their references?
- How do the controllers behave under day-night transitions and disturbances?
- Is the achieved performance obtained with reasonable control effort?

After running `Benchmark_main`, the benchmark provides two complementary outputs:

- A numerical performance summary printed in the MATLAB Command Window
- A set of figures that show the time evolution of the main variables and actuator signals comparing to the baseline controllers.

Some figures examples are displayed in Figures 3 and 4.

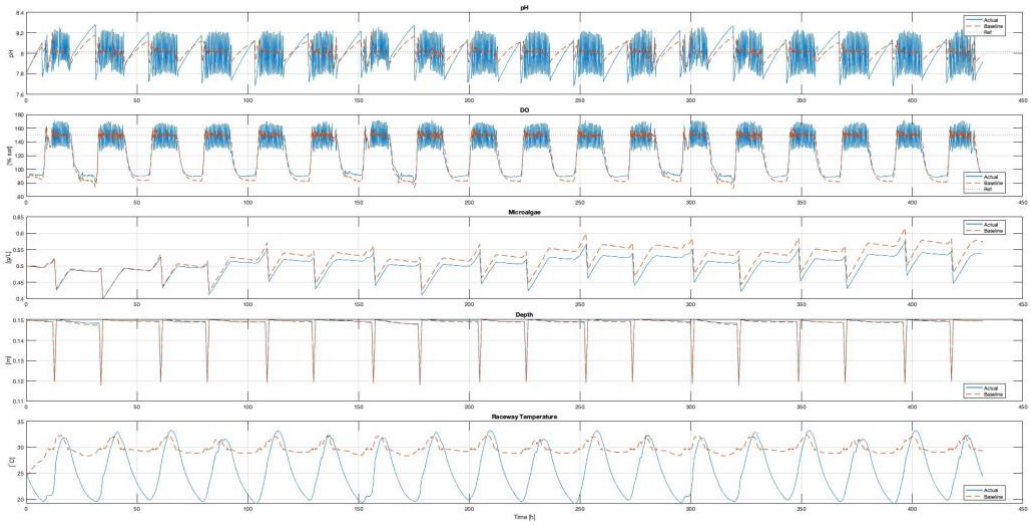


*Figure 3 - Outputs results.*

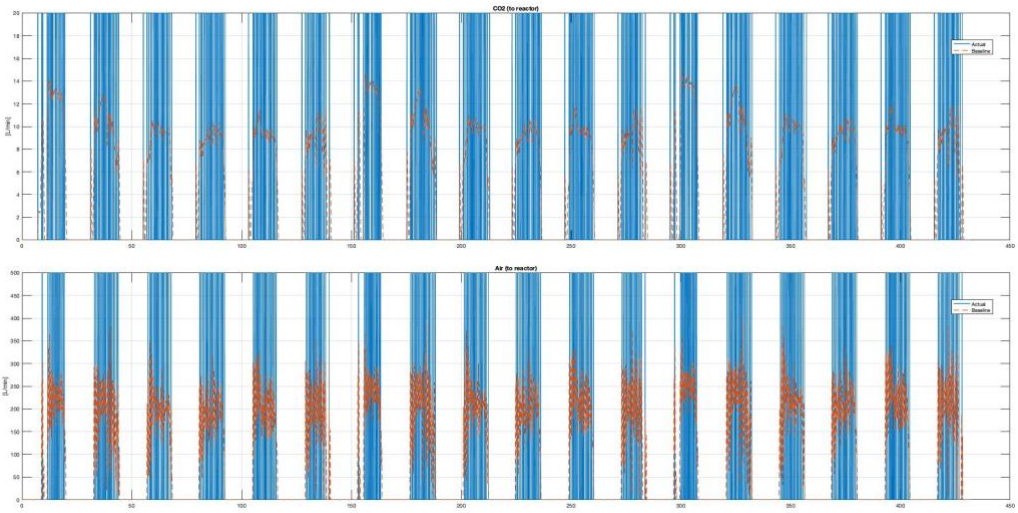


*Figure 4 – Inputs results.*

# Performance summary printed in the Command Window

## Dynamic cost functions (J)

At the end of the simulation, MATLAB prints the dynamic cost breakdown:

- `results.J.pH`: cost associated with pH regulation.
- `results.J.DO`: cost associated with DO regulation.
- `results.J.T`: cost associated with temperature regulation
- `results.J.X`: cost associated with productivity (biomass-related performance).
- `results.J.Total`: global score used for ranking.

This is displayed as:

- A table of individual costs (pH, DO, Temperature, Productivity)
- The total cost

Interpretation for participants: your goal is to reduce `results.J.Total` compared to the baseline controller (baseline score is normalized to 1).

**Lower is better!**

## Key Performance Indicators (KPIs)

The benchmark also prints a set of KPIs that help you interpret performance beyond the score:

**Resource usage**

- `results.total_air_L`: total injected air $L$
- `results.total_CO2_L`: total injected $CO_2$ $L$

**Biomass and productivity**

- `results.prod_g`: total produced biomass $g$
- `results.prod_areal_gm2_day`: areal productivity $g/m^2/day$
- `results.harv_total_g`: total harvested biomass $g$
- `results.harv_frac`: fraction of produced biomass harvested
- `results.harv_prod_areal_gm2_day`: harvested areal productivity $g/m^2/day$
- `results.acumm_rel`: relative biomass accumulation

Tip for participants**:** Two controllers can have similar tracking quality but different KPIs. A "good" controller is typically one that keeps pH/DO/T near references without excessive $CO_2$/air usage, while maintaining strong productivity indicators.

The results are showing when executing code `showResults.m` the end of the simulation.

The Cost Function, separated by term and its total, are displayed with the main Key Performance Indicators, as represented in Figure 5, which corresponds to the results presented in Figures 3 and 4.

```
Simulation finished.

Dynamic cost functions (J) - Actual controllers:
        pH      Disolved oxygen    Temperature   Productivity
    2.4646              0.65645        0.96079        0.15694

   ------------------------------------------------------------
   Total =  3.9249


Dynamic cost functions (J) - Baseline controllers:
        pH      Disolved oxygen    Temperature   Productivity
   0.33333              0.33333            0.5        0.16665

   ------------------------------------------------------------
   Total =       1
```

*Figure 3 - Cost Function displayed in MATLAB workspace.*

# Control Objective and Cost Function

The performance of each control strategy in the benchmark is evaluated using a global cost function, denoted as $J_{\text{Total}}$. This scalar value is the only metric used for ranking and comparison of submitted controllers.

The benchmark is designed so that participants do not need to define or modify the cost function. Instead, the cost is automatically computed by the simulator after each run, based on the closed-loop behavior of the system.

## Normalized global cost

The global cost is normalized with respect to a baseline controller, which is provided in the repository.

The baseline controller has a normalized cost. The total cost $J_{\text{Total}}$is composed of several partial cost terms, each associated with a specific control objective:

$$J_{\text{Total}} = \mathrm{J}_{pH} + \mathrm{J}_{\text{DO}} + \mathrm{J}_{\text{Temp}} - \mathrm{J}_{prod} = 1$$

- pH regulation cost
  Penalizes absolute deviations of pH from its reference value and the control increments of the $CO_2$ variable.

- DO regulation cost
  Penalizes absolute deviations of DO from its reference value and the control increments of the Air injection variable.

- Temperature regulation cost
  Penalizes absolute deviations of the culture temperature from its reference value and the control increments of the heat flow and inlet temperature.

- Productivity-related cost
  Reflects the impact of control actions on biomass production and overall process performance.

These partial costs are reported individually in the MATLAB output as:

- `results.J.pH`
- `results.J.DO`
- `results.J.T`
- `results.J.X`

They are then combined internally to produce the **global cost**:

- `results.J.Total`

Participants do not need to combine these terms manually; the benchmark handles this automatically.

Any controller achieving

$$J_{\text{Total}} < 1$$

is considered an improvement over the baseline. Lower values of $J_{\text{Total}}$ indicate better overall performance. The main objective is to design controllers that reduce $J_{\text{Total}}$ as much as possible, while respecting the benchmark structure and constraints.

## How to interpret the cost

When comparing controllers, participants are encouraged to consider both: The global cost $J_{\text{Total}}$, which determines ranking, and the individual cost components, which explain *why* a controller performs well or poorly.

For example: A controller with excellent tracking but excessive $CO_2$ or air usage may reduce tracking costs but increase productivity-related penalties. A smoother controller may slightly sacrifice tracking accuracy but achieve a better overall score due to improved efficiency.

## Practical guidance for students

- Always compare your results against the baseline controller.
- Use resulting plots to understand the behavior behind the cost values.
- Aim for balanced performance: good tracking, reasonable actuator usage, and stable operation over day-night cycles.

Remember: the goal is not perfect tracking at any cost, but the best overall trade-off, as reflected by a reduced $J_{\text{Total}}$.

# Results Submission

Each team must submit the following items:

1. Controller source files
   MATLAB files containing the implemented controller logic for:
   - pH control,
   - DO control,
   - Temperature control.

Only the controller functions should be submitted. The process model and simulation framework must not be modified.

2. Simulation results file.
   The MATLAB results structure generated by running the benchmark:
   - results

## How to submit

All submission materials must be sent by email to:

joseluis.guzman@ual.es
Before submission, participants should ensure that:

- The simulation runs without errors using Benchmark_main.m,
- The reported results.J.Total value corresponds to the submitted controller files,
- No protected files have been modified.
- The registration deadline is May 15, 2026.
- The submission deadline is July 1, 2026.

## Evaluation procedure

Submitted controllers will be evaluated by the benchmark committee following these steps:

1. The submitted controller files are integrated into the official benchmark version.
2. The benchmark is executed using the standard disturbance scenarios.
3. The global cost $J$_Total is computed.
4. Controllers are ranked according to their normalized global cost.

The ranking is based exclusively on $J_{\text{Total}}$. The baseline controller has a normalized cost of 1, and lower values indicate better performance.

## Reproducibility and compliance

To guarantee fair comparison all simulations are run using the same MATLAB version and benchmark configuration.

Only controller files are allowed to differ between submissions.

Any submission that modifies protected files or violates the controller interface will be disqualified.

Participants are therefore strongly encouraged to test their controllers using a clean copy of the repository before submission.